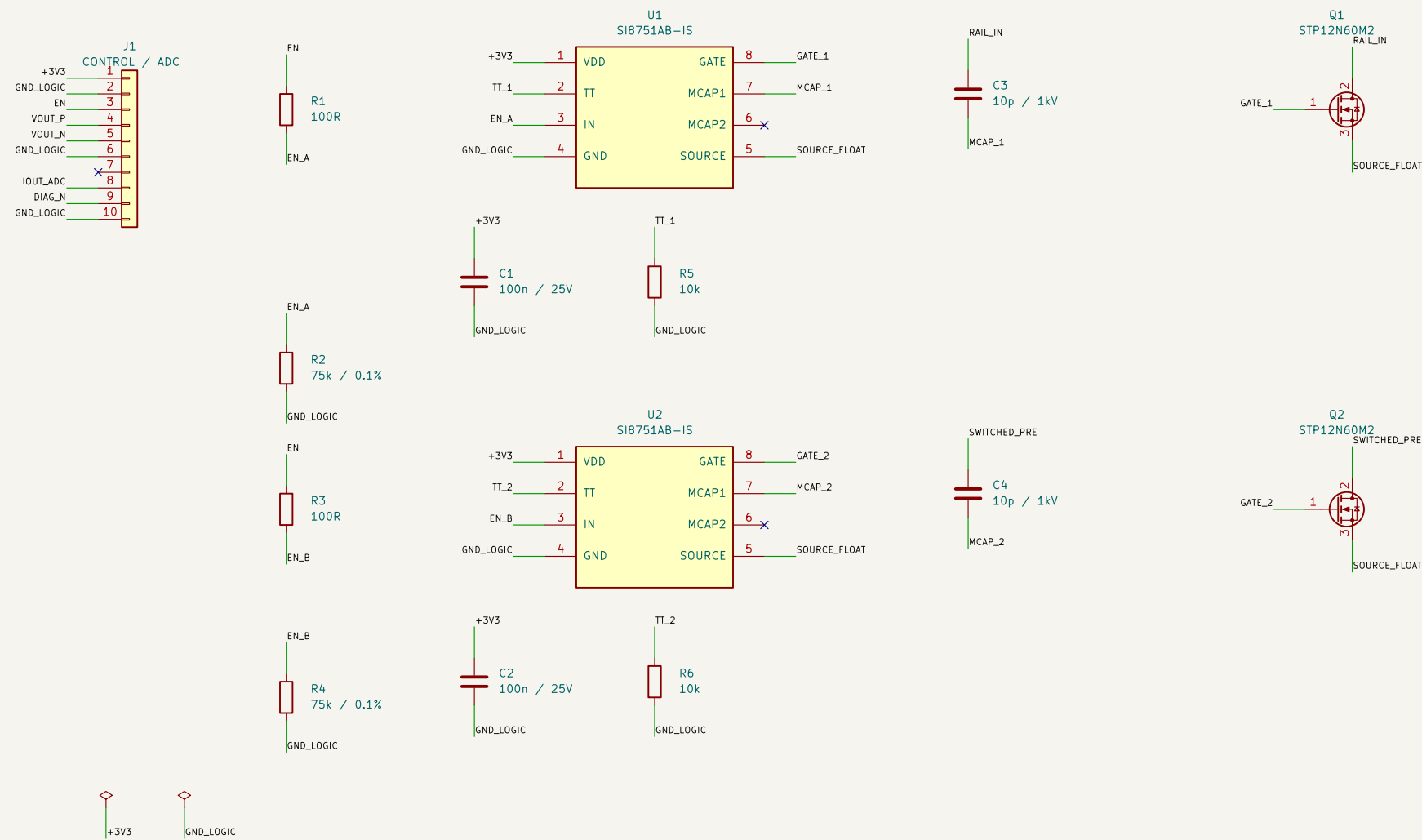


## AC / DC SOLID-STATE SWITCH + ISOLATED VOLTAGE AND CURRENT SENSE

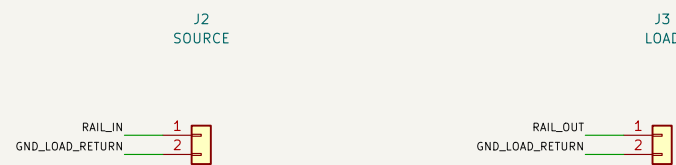
REV C - 120 Vac / 170 Vdc, <1 A - RESISTIVE BENCH LOADS - EXTERNAL FUSE REQUIRED

1. LOGIC CONTROL – 3.3 V supply, active-high EN, both drivers always commanded together



## 2. BIDIRECTIONAL POWER PATH

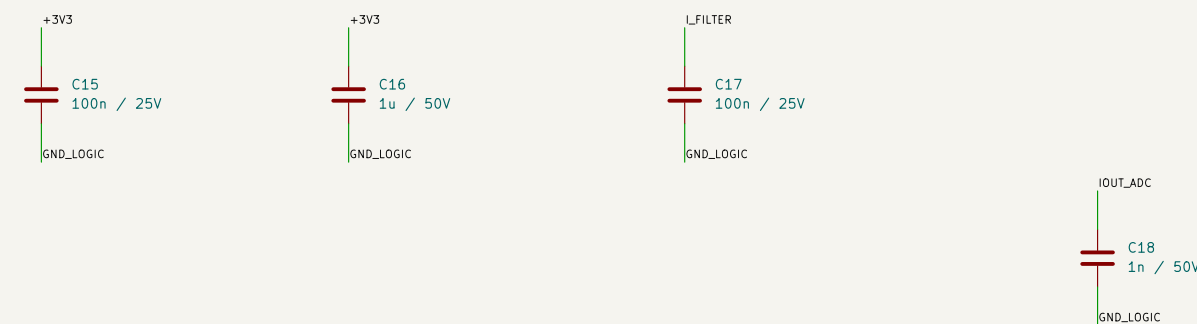
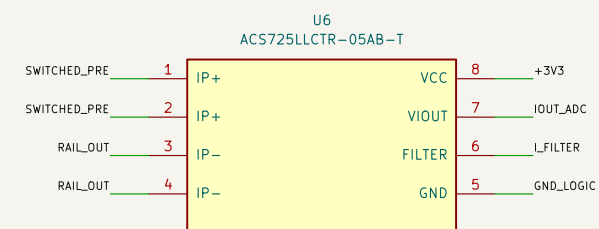
Q1/Q2 sources share SOURCE\_FLOAT.  
Opposed body diodes block either polarity when OFF.  
Gates are separate; driver outputs are not paralleled.  
SOURCE\_FLOAT must NOT connect to GND\_LOAD\_RETURN.



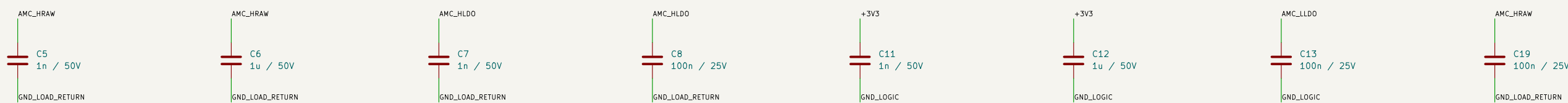
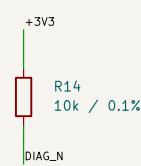
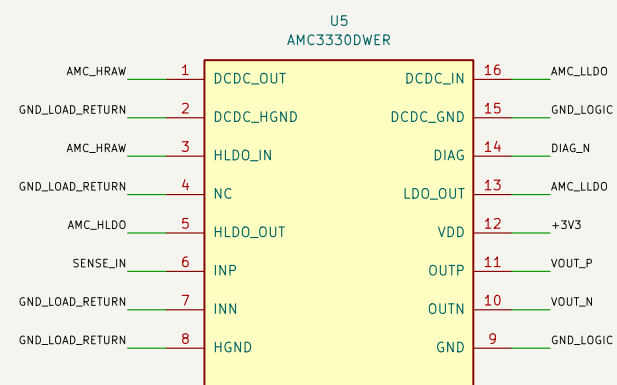
No gate-source bleed resistor: the Si8751 has a high source impedance. A conventional 10k/100k pull-down would collapse its generated gate voltage. Miller clamp connects through the drain capacitor.

Start with resistive loads and  $\leq 1$  on/off cycle per second.  
Inductive loads need separately sized surge suppression.  
Board is a bench prototype; no mains safety certification.  
MOSFET tabs are live and at different potentials.

### 3. ISOLATED CURRENT SENSOR – positive current from Q2 to J3



At VCC=3.3V: IOUT\_ADC =  $1.65V + 0.264V/A * I(A)$ .  
R15/R16 divider removed. Calibrate zero and gain.  
C17=100n gives approximately 884Hz filter bandwidth.



Vrail = 100.5 \* (VOUT\_P - VOUT\_N), with voltages in volts.  
OUTP/OUTN require a differential ADC or external difference amplifier.  
Loss of measurement power is not proof of a de-energized rail.

U3/U4 and C9/C10 retired from revision A. No external isolated converter required.  
J1 pinout changed: pin5 is now OUTN. Supply +3.3V at pin1; pin7 is now NC. Do not apply 5V.  
All resistors and capacitors use 0805 packages. See BOM for inventory matches.